Execution Report #3

The tower, the work unit, and the sealed room

B2 (inspection tower), C0 (work-unit machine), C1 (noninterference) — presented in the program's new expository house style, here inaugurated

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A note on form. The dedicated research channel was unavailable this session, so the expository conversion was done in-house: §1 codifies the house style that all results — these three and everything after — will now follow by default, and §2 retells the prior results in it. A deeper survey of the exposition craft itself (worked-example research, explorable explanations, the writing canon) is queued as a research task at the end of the accompanying message; its findings will refine, not replace, this template.

1 The house style: seven moves

Every result is presented through the same seven moves, in order. The discipline is *intuition before formalism, numbers before generality, boundaries always.*

1. **The problem as a story.** One concrete scene a smart engineer recognizes. No jargon yet.
2. **The idea in one breath.** A single italic sentence carrying the whole result.
3. **Intuition before formalism.** An analogy chosen for *structural* similarity (the relations map), never surface similarity. Terms earn their definitions through use before they are defined.
4. **The box.** The definition or theorem, precisely stated, *after* the reader already believes it. Boxed so it can be cited without the prose.
5. **Numbers you can check by hand.** A worked example small enough to verify mentally or on one line of arithmetic.
6. **What it buys.** The application — which product claim, chapter, or paper this result underwrites.
7. **The honest boundary.** What the result does *not* say, stated as prominently as what it does. A result without a boundary paragraph is not done.

2 The story so far, retold

Six results are already in hand; here is each in capsule form (one breath + one number).

Read-poverty and the floor (A7). A human overseeing a swarm cannot read everything, so agents write summaries — but a summary is a channel, and channels have capacities. **One breath:** *no summary of fewer than* $\log_2 \binom{N}{k} - \log_2 \binom{m}{k}$ *bits can guarantee an operator who opens* m *items catches all* k *dangerous ones among* N*.* The floor survived its own falsification experiment: 0/16 violations, even by an oracle that knows the answer — because the bits are charged for *identifying* the set, and no encoder can smuggle identity past the meter.

One summary cannot serve two readers (A1). The agent that resumes your work wants what is *useful*; the human who oversees it wants what is *dangerous*; a routine-but-essential file and an abandoned experiment with irreversible side effects order oppositely under those two lenses. **One breath:** *a single ranking serves two readers if and only if they agree on every pair — and these two provably do not.* The kicker is quantitative: serving both readers with one zero-miss digest costs not twice the bits but *2.13*×, super-additive, because catching $2k$ needles among N is harder than twice catching k.

The attention formula was right for a derivable reason (A2). The queue ranked items by stakes × irreversibility × anomaly as a design instinct. **One breath:** *that product is exactly the expected unrecoverable loss of looking away — provided "anomaly" is a calibrated probability — and the optimal look/pass rule is the classical likelihood-ratio test.* Corollary with teeth: reputation may enter only through the probability, so no track record can zero out a catastrophic irreversible item.

Summaries have a price curve, and zoom is the discount (B1). Treat digest-making as lossy compression with two dials — misses tolerated, items opened. **One breath:** *the digest rate is a closed-form function $R(\delta, f)$ trading bits against opens against misses, and adaptive drill-down ("zoom") cuts a conservative digest's opens from F to about $k \log_2(F/k)$.* Measured: 15.3× fewer opens at $F{=}2500, k{=}10$ — and honestly, no advantage at all when the flagged set is dense; zoom pays exactly in the safe, over-flagging regime.

What a runtime can prevent is a solved question from 1987 (B3). Which agent policies can be *stopped before effect*, and which only detected after? **One breath:** *exactly the safety policies controllable with respect to the events no mediator can intercept — Ramadge–Wonham supervisory control, with the model's inner life as the uncontrollable alphabet.* The checker's crown jewel: "no egress *after* reading a secret" is preventable, because it permits the uncontrollable read and gates only the controllable egress — the computational proof that clean rooms must gate the channel, never the token.

The sheaf question got an honest answer (the gate). Does cohomology detect a two-faced signer beyond just comparing notes pairwise? **One breath:** *yes — exactly when the missing comparison lies on a cycle, so the sum-around-the-loop constraint substitutes for the comparison you couldn't make; across a cut edge, never.* Two wrong formulations located that boundary, which is the falsification discipline working as designed.

3 B2 — Who audits the auditors, and what it costs

The story. A marketplace where AI judges grade AI work. A judge can take a bribe: pass shoddy work, pocket G. So you audit the judges. But auditors can be bribed too — and now you seem to need watchers all the way up, each level more expensive than the last. Does the tower ever stop?

> *Honesty holds when expected punishment beats the bribe — and the tower of auditors* converges*, because each level has geometrically less corruption left to protect, provided the judges are drawn from enough independent camps that no single deal buys them all.*

Intuition. Traffic enforcement does not put an officer on every corner; it makes *expected fine × odds of being caught* exceed the minutes saved by speeding. That is the stage game. The tower is the traffic *court*: fixing a ticket only pays if the ticket was worth more than the judge's price — and if judges are drawn sealed from eight rival benches, a briber must pay all eight to be safe, which is only worth it for enormous tickets. Small corruption cannot afford its own protection; large corruption buys protection but bleeds value doing so; either way, what survives each level shrinks. Heterogeneity — judges from different principals and different model families — is not a diversity nicety; it is the mechanism that makes “bribe them all” expensive.

Theorem (stage). A judge with one-shot corrupt gain G, facing audit probability ρ, detection probability d, and slashable bond B, is deterred iff $\rho dB \geq G$; the critical committed audit rate is $\rho^\star = G/(dB)$ (Becker’s condition). If capture also confiscates the gain, $\rho^\star_c = G/(d(G{+}B)) < \rho^\star$. In the simultaneous (uncommitted) game the mixed equilibrium is $\rho^\star = G/(dB)$, $q^\star = a/(dL)$.

Theorem (tower contraction). Level $k{+}1$ audits level-k auditors, sampled sealed from a pool spanning C disjoint cliques; an auditor’s bribe floor is its expected forfeiture $\beta = \rho dB$. Rational bribery is all-or-nothing and profitable iff $G_k > C\,B$; below that threshold, corrupt value decays geometrically, $G_{k+1} = (1-\rho d)\,G_k$. Hence *finite* bond capital (B per level, $\lceil \log G_0 / \log \frac{1}{1-\rho d} \rceil$ levels) certifies a tower of unbounded depth.

Theorem (amortization). As a judge’s verified history t grows, reputation-at-stake vt grows, and incentive-compatible audit schedules decline: with loss only-if-audited, $\rho_t = G/(d(B+vt))$ and lifetime spend is $\Theta(\log T)$; with cheats also surfacing later at rate r independent of audits, $\rho_t = \max\bigl(0, \frac{G-rvt}{dB}\bigr)$ reaches zero at $t^\star = G/(rv)$ and lifetime spend is $O(1) = aG^2/(2dBrv)$. Flat auditing spends $\Theta(T)$. *Reputation is amortized verification.*

Numbers. G=10, d=0.8, B=50: $\rho^\star = 10/40 = 0.25$ — audit one grade in four, forever, and cheating exactly stops paying (verified: cheat payoff $= +0.000000$ at $\rho^\star$). Tower with G_0=400: at C=1, bribery keeps corruption on life support ($400 \to 390 \to 380 \ldots$, bleeding only the bribe price per level); at C=8, bribery is uneconomical from level one and value collapses geometrically at 0.8^k ($400 \to 320 \to 256 \ldots$). Amortization at T=200: flat spends 50.0; the log schedule 25.6 (closed form 25.5); the $O(1)$ schedule 35.1, saturating at $aG^2/(2dBrv) = 41.7$. Both schedules pass the pointwise deviation check: maximum deviation value ≤ 0 up to float epsilon.

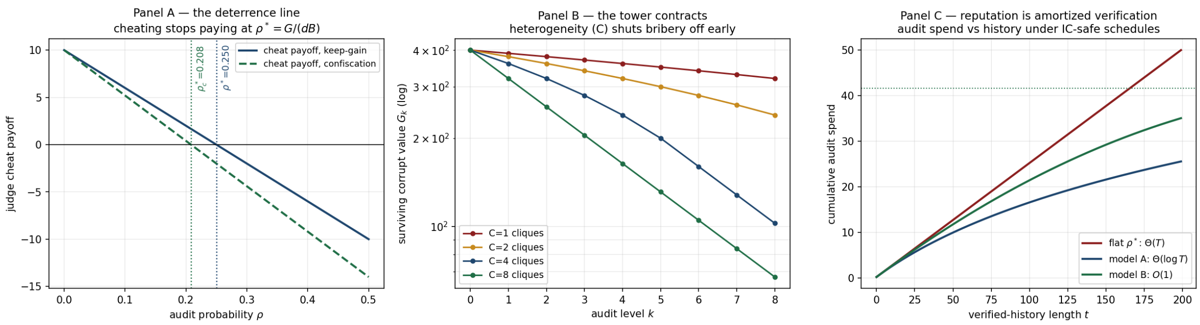

Figure 1: **A:** the deterrence line — cheating’s payoff crosses zero at $\rho^\star$, earlier under confiscation. **B:** the tower — corrupt value per audit level (log scale); more cliques C means bribery shuts off earlier and decay goes geometric. **C:** cumulative audit spend — flat is linear forever; history-aware schedules bend to $\Theta(\log T)$ and $O(1)$.

What it buys. Conjecture III.11.1 — the volume’s central open question — becomes a parameterized theorem with named sufficient conditions and a measured budget. It also fuses the two external reviews’ proposals into one machine: *model heterogeneity* is where the C disjoint cliques come from, and *VRF honeypots* are the implementation of the sealed random audit at rate $\rho^\star$. Neither review had the theorem; together they were it. This is Paper 3’s core, now executed.

Honest boundary. $\rho^\star$ depends on the payoff convention (keep-gain vs. confiscation — state which your settlement rule implements). The tower needs an exogenously honest root (the operator at n=1; a bonded arbitration market at scale) and *sealed* sampling — a briber who learns the draw beats the clique math. Detection d and cross-clique collusion correlation are measured quantities, not assumptions; and whether amortization is $\Theta(\log T)$ or $O(1)$ turns on an empirical fact — do cheats surface without audits? — that deployment telemetry must answer.

4 C0 — The work unit: a case file that outlives every clerk

The story. An agent dies mid-task. Its successor — maybe a different model from a different vendor — picks up and later says "done." Who checks the paperwork? Where is the record that the port it held was released, that the money moved once and not twice, that "verified" meant an inspection actually happened? The answer the whole program converges on: a durable *work unit* — a case file that outlives every clerk who touches it.

Make the unit of trust a work unit whose every consequential step passes a guard, and then prove*, by exhausting the machine's reachable states, that its six promises can never be broken — and that every guard is load-bearing, because removing any one lets the checker construct the crime.*

Intuition. A notarized escrow folder. Nothing enters without the right stamp (an active grant carrying the *current* role epoch — so a clerk who was replaced yesterday finds today's stamp rejects him); no payment leaves twice (each external operation carries an idempotency key, honored at most once); and "verified" is not a mood — it means the acceptance policy and the inspector's receipt are physically in the folder. We then do something a paper folder cannot: enumerate *every state the machine can ever reach* and check all six promises in each one.

The machine. State = (work-unit phase, policy, receipt, role epoch + holder + stale token, grants with parentage, effect journal, idempotency journal, settlement). Transitions: propose, authorize, start, admit-effect (guarded on current epoch), reassign-role (epoch++, old holder keeps a stale token), delegate (guarded on scope $\subseteq$ parent), external-op (guarded at-most-once), witness, admit-receipt, verify (guarded on policy $\wedge$ receipt), contest, settle (guarded on owner-funded $\wedge$ verified $\wedge$ receipted).

Result. On the two-principal instance, **all 536 reachable states satisfy all six invariants** (stale-epoch exclusion, at-most-once settlement, evidence-backed verification, capability attenuation, fencing at the effect boundary, correctly-funded settlement). **Mutation suite:** disabling any single guard yields a violation, found automatically as a *shortest counterexample trace*; restoring guards restores safety.

Numbers. The five crimes, each with its minimal script: the stale write in **4 steps** (authorize → start → reassign → the old holder's ghost writes with yesterday's epoch — precisely the "paused holder wakes after expiry" scenario the rigor review warned about); the double-settle in **2**; the hollow "verified" in **4**; the escalated delegation in **1**; the wrong-principal payout in **7** — notably, the checker had to walk the entire *legitimate* path (policy, receipt, witness, verify) before the one illegal step, which is exactly how that fraud would look in production.

What it buys. This is C0 — the formal spine of the re-spined volume (Document 1, Move 1) and the Phase-1 product's promises turned into checked properties: the work-unit receipt, at-most-once external effects, fenced role leases, attenuated delegation. Every earlier theorem now has a substrate to be a property *of.* The unbounded-parameter version is the TLA$^+$/Apalache obligation; this executable spec and its mutation suite are what certify the spec is not vacuous before that investment.

Honest boundary. Bounded model checking on a small instance: it proves the *spec*, at these parameters, not the deployed daemon — deployment must *refine* this machine (every real code path maps to a transition), which is a separate obligation. Safety only: liveness ("the work eventually settles") needs fairness assumptions this check does not make. And an invariant list is only as good as its coverage; the mutation suite certifies these six have teeth, not that they are complete.

5 C1 — The sealed room with one mail slot

The story. Derek has the spreadsheets; Erin has the model. Neither will hand over the crown jewels, yet both want the answer. So: a sealed room. Erin's agent works inside; the only opening in the wall is a mail slot — the declassification gate — through which a *declared* summary may pass. The question a CISO will ask: how do you know the wall has no other holes?

Run the whole world twice — identical except for Derek's secret — under every possible *sequence of moves, and prove Erin's view is bit-for-bit identical whenever the gate's declared function maps the two secrets to the same release.*

Intuition. You cannot inspect a wall for holes by staring at bricks; you inspect it by standing outside in two parallel worlds and asking whether anything you can see ever differs. If Derek's secret is 0 in one world and 2 in the other, and the gate is contracted to release only *parity*, then a sound room makes those worlds indistinguishable to Erin forever — under every interleaving of loads, reads, submissions, releases, and reports. That is noninterference *modulo declassification*: silence except through the slot, and through the slot, only what the contract says.

Property (NI mod declassification). For secrets s, s' and every action sequence: if $g(s) = g(s')$ then Erin's observation sequences are identical; if $g(s) \neq g(s')$ they may differ only at explicit gate-release steps. Here $g(s) = s \bmod 2$ is the contracted release.

Result. Verified **exhaustively** on the finite clean-room model: all interleavings to depth 7 × all secret pairs — the property holds, the schedule itself is secret-independent (checked separately, since an enabledness difference would itself be a channel), and the unwinding condition *local respect* (Derek-side actions never touch Erin-observable state) holds mechanically. **Mutations:** (m1) a gate that leaks the raw secret is caught distinguishing secrets $(0, 2)$ — an *equal-parity* pair the honest gate provably keeps identical; (m2) a worker bypassing the gate is caught in a 3-step trace. Reverting mutations restores the property.

Numbers. Four secrets, seven actions deep, every interleaving, every pair: zero distinguishing observations in the honest model. The m1 witness in full: `load` → `read` → `submit` → `release`, after which Erin sees `(gate,0)` in one world and `(gate,2)` in the other — a four-step, human-readable proof that this particular gate breaks its contract.

What it buys. The airlock's headline claim, verified from a second independent direction: B3 proved *by controllability* that the clean room must gate the channel and not the token; C1 now proves *by two-run equivalence* that gating the channel suffices for the contracted secrecy. Together they underwrite the corrected product sentence — *every release is explicit, gated, and bounded* — and they are Paper 4's finite-model core.

Honest boundary. Finite depth, finite secrets: the general theorem over unbounded state is the Isabelle/AFP unwinding obligation (Rushby), for which this model is the blueprint. The guarantee is possibilistic — timing, token counts, and termination channels are out of model, as the design declares. The gate's *specification* is the residual oracle: we prove releases match g; choosing a g that leaks too much is a contract failure no theorem prevents.

And a human who reads a released parity may still remember it — the system bounds channels, not minds.

6 Program status

Item	Status	Headline	Feeds
A1, A2, A7, B1	done	Paper 1 complete (floor, split, policy, frontier)	Paper 1
B3	done	regimentable = controllable; channel-not-token	Papers 2, 4
Sheaf gate	decided	conditional commit; cycle boundary	Paper 7
B2	**done**	$\rho^{\star}$; tower contracts; amortized verification	Paper 3
C0	**done (bounded)**	536 states safe; 5/5 mutations caught	spine; Paper 8
C1	**done (finite)**	NI mod declassification, exhaustive; 2/2 mutations	Paper 4

Remaining from the roadmap: the Tier-A quick wins not yet formalized (A3 ε-ledger, A4 canaries, A6 no-mint), the Chapter-III pair (B5 engine substitution, B6 probation cliff), B4 (the tractable conflict fragment), C2 (minimum viable take-rate), and the lifting of C0/C1 to unbounded form (Apalache; Isabelle). All future write-ups follow §1.

Reproducibility. Three self-contained scripts (`b2_tower.py`, `c0_workunit.py`, `c1_noninterference.py`); every number above regenerates deterministically, and every mutation prints its witness trace.